Solving the static Einstein equations with physics-informed neural networks

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(Dated: October 28, 2026)

A classical existence theory guarantees that the stationary vacuum Einstein equations, posed in the region outside a ball, have a unique solution for boundary data close enough to flat space, depending continuously on the data. We present a physics-informed neural network (PINN) solver for this boundary-value problem in the static case. The unknowns are a Riemannian 3-metric h and the squared norm λ of the static Killing vector, in harmonic gauge; the data are λ and the induced metric on an inner sphere, and asymptotic flatness is imposed by Robin conditions on an outer sphere. The connection is the Christoffel symbol of the network metric. We test the solver on Schwarzschild in harmonic coordinates and on two-rod Weyl solutions, on the symmetry axis and rotated so that no symmetry is left. For quadrupolar data, which have no closed-form solution, we compare with an independent spectral solution of the same problem. The harmonic gauge and the inner data leave the asymptotic chart free to translate, and this translation dominates the error until it is fixed by Cederbaum's pseudo-Newtonian centre of mass, a conserved flux of the static equations; with it, the solution agrees with the spectral one to five to six orders below the size of the perturbation. Finally, we read the Geroch–Hansen mass moments of the computed solutions off the symmetry axis, without constructing Weyl coordinates; they reproduce the exact moments of the Weyl solutions and the reference moments of the quadrupolar one.

I. INTRODUCTION

For many years there has been a mathematical theory of the stationary Einstein equations in an asymptotic region of spacetime, namely the region outside a ball [1]. It states that boundary data close enough to those of flat spacetime, together with the asymptotic conditions, produce a unique solution that depends continuously on the data. Very few attempts have been made to explore this theory numerically [2]. Here we show that this is possible, and that it opens a new window on these spacetimes.

The method also gives a simple way to explore other systems of possible physical importance: systems in which one solves for a Riemannian metric together with other fields, through equations that contain its Ricci tensor. Examples are the construction of minimal-energy data sets [3] and the equations that give higher-order Newtonian-limit expansions [4]. [TODO: more examples]

We restrict this work to the static Einstein equations. The stationary case is no harder from the point of view of the approximation method, and is left for a continuation of this work.

Physics-informed neural networks [5] represent the unknown fields by a neural network and minimise the squared residual of the equations and boundary conditions over sampled points. They need no mesh, handle the non-compact exterior through a logarithmic radial variable, and treat a configuration with no symmetry exactly as they treat a symmetric one. A small residual does not by itself mean a small error, so every result below is graded against an exact solution or an independent reference.

Section II states the equations, the gauge, the boundary and asymptotic conditions, and the residual gauge freedom. Section III describes the solver. Section IV tests it on exact solutions, Sec. V solves the quadrupolar problem, and Sec. VI extracts the multipole moments. Appendix A lists what is needed to repeat every computation.

II. THE STATIC EINSTEIN EQUATIONS

A. Variables and field equations

We write a static vacuum spacetime as

$$g = -\lambda\,\mathrm{d}t^2 + \lambda^{-1}h, \tag{1}$$

where h is a Riemannian metric on the spatial slice and $\lambda = -\xi\cdot\xi$ is the squared norm of the static Killing vector $\xi = \partial_t$; if $g = -N^2\mathrm{d}t^2 + \gamma$, then $\lambda = N^2$ and $h = N^2\gamma$. The vacuum equations become

$$R_{ab}(h) = \frac{1}{2\lambda^2}\,\nabla_a\lambda\,\nabla_b\lambda, \tag{2}$$

$$h^{ab}\nabla_a\nabla_b\lambda = \frac{1}{\lambda}\,h^{ab}\nabla_a\lambda\nabla_b\lambda. \tag{3}$$

With $U = \frac{1}{2}\ln\lambda$ these read $\mathrm{Ric}(h) = 2\,\mathrm{d}U\otimes\mathrm{d}U$ and $\Delta_h U = 0$.

In some cases we shall solve for the first system of equations, in some cases for the second equivalent system, depending on the problem. The difference being that U is closer to the Newtonian potential, and λ is the squared norm of the Killing vector. Thus, for instance the first multipole moments of U are closer to the Newtonian ones. There are differences, ofcourse, and in fact one finds the $\frac{1}{\rho^2}$ term in λ but not in U.

B. Gauge

We impose harmonic coordinates on h,

$$\Gamma^i \equiv h^{jk}\Gamma^i{}_{jk} = S^i, \tag{4}$$

with $S^i = 0$; equivalently $\Delta_h x^i = 0$ for the Cartesian-like coordinates $x^i = \rho\, n^i$. For the Weyl tests we instead use a source built from the candidate's own metric,

$$S^i = (h_{\rho\rho} - 1)\, h^{ij}\partial_j \ln \rho_n, \qquad \rho_n = \big|x - (x\cdot n)\, n\big|, \tag{5}$$

with n the symmetry axis, which Weyl's canonical form $h = e^{2k}(\mathrm{d}\rho^2 + \mathrm{d}z^2) + \rho^2\mathrm{d}\phi^2$ satisfies exactly.

C. Test solutions

There are known solutions for this system, we shall use some of them to test our solver.

1. *The exact spherical family*

In the harmonic chart the spherically symmetric solutions are

$$h_{ij} = \Big(1 - \frac{R_0^2}{\rho^2}\Big)\delta_{ij} + \frac{R_0^2}{\rho^2}\, n_i n_j, \qquad \lambda = k\,\frac{\rho - R_0}{\rho + R_0}, \tag{6}$$

i.e. $h = \mathrm{d}\rho^2 + (\rho^2 - R_0^2)\,\mathrm{d}\Omega^2$. With $k = 1$ this is Schwarzschild in harmonic coordinates: $M = R_0$, areal radius $r = \rho + R_0$, $\lambda = 1 - 2M/r$. The harmonic condition still allows the radial maps $\rho \to c_1\rho + c_2 F_2(\rho)$, $F_2 = 2R_0 + \rho \ln\frac{\rho - R_0}{\rho + R_0}$, fixed by the round inner metric and $h \to \delta$.

2. *Weyl two-rod solutions*

Static axisymmetric vacuum metrics can be written in Weyl's canonical form [6],

$$g = -e^{2U}\mathrm{d}t^2 + e^{-2U}\big[e^{2k}(\mathrm{d}\rho_c^2 + \mathrm{d}z^2) + \rho_c^2\mathrm{d}\phi^2\big], \tag{7}$$

where ρ_c is the cylindrical radius, U is an axisymmetric solution of the flat Laplace equation, and k follows from U by a line integral, $\partial_{\rho_c} k = \rho_c\,[(\partial_{\rho_c}U)^2 - (\partial_z U)^2]$, $\partial_z k = 2\rho_c\,\partial_{\rho_c}U\,\partial_z U$, with $k \to 0$ at infinity. In our variables $\lambda = e^{2U}$ and $h = e^{2k}(\mathrm{d}\rho_c^2 + \mathrm{d}z^2) + \rho_c^2\mathrm{d}\phi^2$. A rod of length L on the axis, $U_{\mathrm{rod}} = \frac{1}{2}\ln[(R_a + R_b - L)/(R_a + R_b + L)]$ with R_a, R_b the distances to its ends, is the Schwarzschild solution of mass $L/2$; two rods give two black holes on the axis (held apart by a strut, where $k \neq 0$ between them). We evaluate k by Gauss–Legendre quadrature. The Weyl chart is not harmonic, but it satisfies the gauge condition (4) with the source (5), which is what we impose in these tests. The inner boundary is a coordinate sphere $\rho = \rho_{\mathrm{in}}$ large enough to enclose both rods, and the data on it, λ and the induced metric, are those of the exact solution.

To test a configuration with no symmetry we rotate the solution by an angle α about the y axis. The equations are covariant, so $h \to R\,h\,R^T$ and $\lambda \to \lambda$, evaluated at $R^T x$, is again a solution. The inner data are taken from the rotated fields, and the gauge source (5) is taken about the rotated axis $n = R\hat{z}$: the gauge condition is a statement about the chart and rotates with the configuration.

D. Boundary and asymptotic conditions

The domain is the shell $\rho_{\mathrm{in}} \le \rho \le \rho_{\mathrm{out}}$.

a. Inner sphere. We prescribe λ and the induced metric,

$$\lambda = \lambda_0 + S_1\, n\cdot e + S_2\, P_2(n\cdot e), \qquad PhP = \frac{r_{\mathrm{in}}^2}{\rho_{\mathrm{in}}^2}\,P, \tag{8}$$

with $P = I - n\otimes n$, r_{in} the areal radius of the sphere in h and e the data axis. No condition is imposed on h_{rr} or h_{rA}. In harmonic gauge these are determined by the solution itself, and prescribing them would leave no solution of the boundary-value problem except when the prescribed values happen to coincide with the solution's own, which are not known in advance. Numerically this shows up as a loss that stalls instead of decreasing: the spectral solution, with $h_{rA} = 0$ imposed, stopped at a residual of 1.8×10^{-5} with a non-decaying $l = 2$ part, and reached 3×10^{-9} once the condition was removed. λ_0 is chosen so that $\lambda \to 1$ on the spherical branch. In the traditional approach to the problem one would look for boundary conditions that would imply the enforcement of the harmonic gauge condition once the equations are solved. In our approach the harmonic gauge condition is imposed directly in the interior: since the connection is the Christoffel symbol of the network metric, $\Gamma^i = S^i$ is a condition on the first derivatives of h, and it enters the loss as one more residual, together with the field equations (Sec. III); the boundary conditions then only have to select a unique solution among those of the equations and the gauge condition. This considerably simplifies their choice, since no conditions are needed whose role is to propagate the gauge.

b. Outer sphere. Asymptotic flatness is imposed by Robin conditions of order n,

$$B_n[f] \equiv \prod_{i=0}^{n-1}\big(\rho\,\partial_\rho + p_f + i\big)\,(f - f_\infty) = 0, \tag{9}$$

which annihilate the decaying powers $\rho^{-p_f}, \ldots, \rho^{-p_f - n + 1}$. The exponents follow from the equations, $\delta h \sim \rho^{-2}$, $\Gamma \sim \rho^{-3}$, $\lambda - 1 \sim \rho^{-1}$; the exact solution (6) gives local decay rates 2.00, 3.005 and 0.952 at $\rho = 20$. For the scalar we impose the condition

on the Geroch–Hansen potential [7, 8]

$$\mathcal{P} = \frac{1-\lambda^2}{4\lambda} = -\tfrac{1}{2}\sinh 2U, \tag{10}$$

whose monopole $M/\rho + M^3/\rho^3 + \dots$ has no ρ^{-2} term, unlike $\lambda = 1 - 2M/\rho + 2M^2/\rho^2 - \dots$. $\lambda_\infty = 1$ is fixed. These pointwise conditions on the outer sphere are all that is imposed there. We also tried strengthening them: the same conditions on further spheres inside ρ_{out} (at $0.7, 0.8, 0.9\,\rho_{\text{out}}$), and spherical means of order-one Robin combinations of h_{rr}, $\operatorname{tr} h$ and $\mathcal{P}$ at ρ_{out}. They were introduced while the inner condition still prescribed h_{rA}, to stop a drift of the far field; with the inner condition (8) and the centre of mass fixed they do not improve the solution (Sec. V C), and we do not use them.

E. The residual gauge freedom and the centre of mass

The harmonic condition fixes the chart only up to harmonic diffeomorphisms $x \to x + \xi$, $\Delta_h \xi = 0$. One of them is left free by our data: $\xi = 0$ on the inner sphere and $\xi \to a$ at infinity. It keeps the chart harmonic and the inner data unchanged, and translates the asymptotic chart by a. Its effect on the scalar, $\xi \cdot \nabla\lambda$, vanishes at ρ_{in} and decays like ρ^{-2}, so the outer conditions barely penalise it: for $|a| \approx 0.04$ it changes the dipole of $\mathcal{P}$ at $\rho = 100$ by $\approx 2.5 \times 10^{-6}$.

It is fixed by the centre of mass. Since U and the coordinates x^i are both h-harmonic, Green's identity makes

$$D^i(S) = \oint_S \left(x^i\, \partial_\nu U - U\, \partial_\nu x^i\right) \mathrm{d}A_h \tag{11}$$

the same on every closed surface S enclosing the inner sphere, and $D = 4\pi M a$ for a mass M displaced by a. This is the pseudo-Newtonian centre of mass that Cederbaum introduced for static ("geometrostatic") isolated systems [9], written in our variables. It is quasi-local, obeys an addition theorem for several separated sources, coincides with the ADM and constant-mean-curvature centres of mass in suitable asymptotic coordinates, and converges to the Newtonian centre of mass in the Newtonian limit [9, 10]. The asymptotic notions it is compared with go back to Regge–Teitelboim and Beig–Ó Murchadha [11, 12] and to the foliation-based definitions of Huisken–Yau, Metzger, Huang, Corvino–Wu and Eichmair–Metzger [13–18] (see Ref. [19]); Cederbaum's analysis builds on the asymptotics of static fields of Beig, Beig–Simon and Kennefick–Ó Murchadha [20–22]. We impose $d \equiv D/4\pi = 0$ on a set of spheres. This is a gauge choice, available for any data; Sec. V shows that without it the translation mode is the largest error of the computed solution.

III. THE SOLVER

a. Networks. The inputs are $n = x/\rho$ and $t = \ln(\rho/\rho_{\text{in}})/\ln(\rho_{\text{out}}/\rho_{\text{in}}) \in [0,1]$, optionally with ρ_{in}/ρ; the body is a tanh multilayer perceptron, the output layer is initialised small so that training starts near flat space. The network outputs h and $\lambda = e^u$, and the connection is the Christoffel symbol of h, computed by automatic differentiation, so the compatibility equation holds identically. (A first-order variant outputs Γ as well and adds compatibility as a residual; on the same Schwarzschild problem it is three orders of magnitude less accurate, Sec. IV B, and it must never be trained with a vanishing compatibility weight, which the code refuses.) For axisymmetric problems there is a simpler system, namely we look for h and λ of the form,

$$h_{ij} = (1+a)\,\delta_{ij} + b\,n_i n_j + \tfrac{c}{2}\,(n_i z_j + z_i n_j) + d\,z_i z_j, \qquad \lambda = e^u, \tag{12}$$

with a, b, c, d, u functions of $(t, \mu = n\cdot z)$: 2285 parameters at width 20 and depth 6 with the decay feature (2265 without). The general three-dimensional network of the same size has 2347. Spherically symmetric tests use $h = \alpha(\rho)\delta + \beta(\rho) nn$.

b. Loss. With R_g the residual of group g (Ricci, gauge, scalar equation), the loss is

$$L = \sum_g w_g\, \overline{(\rho^{p_g} R_g)^2} + w_{\text{in}}\overline{r^2_{\text{in}}} + w_{\text{out}}\overline{r^2_{\text{out}}} + w_{\text{com}} \sum_S |d|^2, \tag{13}$$

with means over the sampled points. $p = (2,1,2)$ makes every residual dimensionless; for the quadrupolar problem we use $p = (4,3,4)$, an extra weight ρ^4 that determines the far field. $w_{\text{in}} = w_{\text{out}} = w_{\text{com}} = 100$ (10 for the Weyl tests). The centre-of-mass term uses a fixed product quadrature on each sphere, Gauss–Legendre in $\mu = \cos\theta$ and uniform in ϕ: 24×8 nodes for the axisymmetric network and 48×32 for the three-dimensional one (with data about an axis other than z, the coarser rule aliases the angular structure on the inner sphere; 48×32 agrees with 96×64 there).

c. Sampling points Interior points are drawn log-uniformly in ρ with uniform directions, boundary points uniformly on the inner and outer spheres; all are redrawn every 500 iterations. Every result is checked on five fresh samples, on which the loss agrees with the training loss to within a few per cent (the ratio is 0.99–1.05 for the quadrupolar runs).

d. Training. Adam [23] for 5000 steps, then the self-scaled Broyden quasi-Newton method SSBroyden [24, 25] with a dense inverse Hessian carried between blocks of 250 iterations, a strong-Wolfe line search (Armijo backtracking in the tests of Sec. IV) and scale-invariant curvature guards, in stages of 10000 iterations; a block whose result is not finite is rejected. For the quadrupolar problem with the axisymmetric network, Adam (3 minutes)

and two such stages (11 minutes each) reach a loss of 1.3×10^{-11}. Everything runs in double precision with JAX [26] on NVIDIA A30 GPUs; one value and gradient at 16384 points costs 0.048 s.

e. A spectral reference. For the quadrupolar problem we built an independent approximation: the spherical background (6) plus an axisymmetric, reflection-even perturbation, with Legendre modes $l \leq 10$ in the angle and 36 Chebyshev modes in $\ln \rho$ for each of the five metric functions and λ (1008 unknowns), solved by Gauss–Newton on exactly the residual vector of the loss (13). Its loss is 2.8×10^{-11}, its $l = 2$ tail decays as $\rho^{-3.00}$ from $\rho = 20$ to 100, and evaluated with a network run's own loss on that run's samples it gives 2.8×10^{-11} as well: it is a solution of the same discrete problem, obtained by an unrelated method. Its rotated version grades the off-axis runs.

IV. TESTS WITH EXACT SOLUTIONS

A. The pipeline on exact solutions

Before training anything we check the implementation itself: an exact solution is fed to the loss in place of the network, through the same code (the same derivatives, residuals, boundary terms and weights), and every term must vanish to rounding. For (6) the total loss is below 10^{-16}. For the two-rod Weyl solution it is 1.3×10^{-28} with the gauge source (5); with the harmonic condition ($S = 0$) it is not small, since the Weyl chart is not harmonic, which shows that the check is sensitive to the gauge term. The rotated Weyl solution solves the system to 3×10^{-14} or better at every residual. (The averaged outer conditions of Sec. II D that we tried and dropped are not satisfied by these solutions, at the 10^{-9} level, which is one more reason not to impose them.)

B. Schwarzschild in harmonic coordinates

The spherical network with $R_0 = 1$, $\lambda_0 = 0.381966$ on the sphere of h-areal radius 2 and Robin conditions at $\rho_{\text{out}} = 100$ recovers the exact family. The Robin condition is not set to zero, but rather to the value the exact solutions would produce, so it has a source term. Table I compares the metric-only and first-order formulations on the problem. The metric-only network is more accurate by three orders of magnitude (a factor 2600 in h and 1300 in λ). The first-order network's error is a gauge mode: along it the compatibility residual is nearly flat, and an earlier, less converged first-order run found the same geometry in a different harmonic chart, with h_{rr} off by a third. Deriving Γ from h removes the mode. With the homogeneous Robin conditions (no source), the metric-only network reaches a loss of 2.4×10^{-11} and maximum errors of 7.0×10^{-6} in h, 6.0×10^{-6} in Γ and 6.3×10^{-7} in λ against the exact solution, and its mass $M_0 = 0.999997$ (Sec. VI) against $R_0 = 1$; each run takes 9–11 minutes on one GPU. (The centre-of-mass condition plays no role here: a spherically symmetric network has no dipole, and the first-order network's error is a radial chart mode.)

TABLE I. First-order against metric-only network on the same Schwarzschild problem (manufactured Robin source, $\rho_{\text{out}} = 100$, 20000 Adam + 2000 SSBroyden iterations; `Paper/commands_schwarzschild.sh`).

	first order (h, Γ, λ)	metric only (h, λ)
final loss	1.5×10^{-9}	2.4×10^{-11}
$\max \lvert\delta h\rvert$	1.7×10^{-2}	6.5×10^{-6}
$\max \lvert\delta \Gamma\rvert$	8.6×10^{-3}	7.9×10^{-6}
$\max \lvert\delta \lambda\rvert$	1.3×10^{-3}	1.0×10^{-6}

TABLE II. Weyl rods (masses 1 and 0.1), inner sphere moved towards the rods; 3000 SSBroyden iterations, standard units.

shell	dist. to horizon	$R_{ab}R^{ab}(\rho_{\text{in}})$	$\max \lvert\delta h\rvert$	loss
$[7.5, 75]$	5.0	3.3×10^{-6}	3.4×10^{-7}	2.4×10^{-14}
$[4, 40]$	1.5	5.8×10^{-3}	1.5×10^{-5}	7.1×10^{-13}
$[3.5, 35]$	1.0	5.3×10^{-2}	7.3×10^{-5}	3.1×10^{-12}

C. Weyl two-rod solutions

The Weyl class [6] gives exact static axisymmetric solutions with nontrivial multipoles (Sec. II C 2). The inner data, on a coordinate sphere three times the rods' extent for the standard shell, and the Robin sources are taken from the exact solution, and the gauge is (5). For two equal rods (masses 1, gap 1) on the shell $[7.5, 75]$ the axisymmetric network trained with SSBroyden from a random initialisation reaches a loss of 2.06×10^{-13} in 3000 iterations, with maximum errors 6.7×10^{-7} in h and 9.0×10^{-8} in λ (Fig. 1). Unequal rods (masses 1 and 0.1) give 2.4×10^{-14}, 3.4×10^{-7} and 5.2×10^{-8}. Their moments are in Sec. VI. Moving the inner sphere towards the rods raises the curvature on it by four orders of magnitude, and the error with it (Table II).

D. A rotated Weyl configuration

Rotating the closest configuration by 45° removes every symmetry. The equations are covariant, so the rotated fields are again a solution, but the gauge is a statement about a chart and the source (5) must be taken about the rotated axis $n = R\hat{z}$ (Sec. II C 2): with that source the exact rotated solution satisfies the gauge condition to 2×10^{-15}, while it violates the harmonic one ($S = 0$) by $|\Gamma^i| \approx 2.6 \times 10^{-2}$. The general three-dimensional metric-only network, with 8192 collocation points and SSBroyden from a random initial-

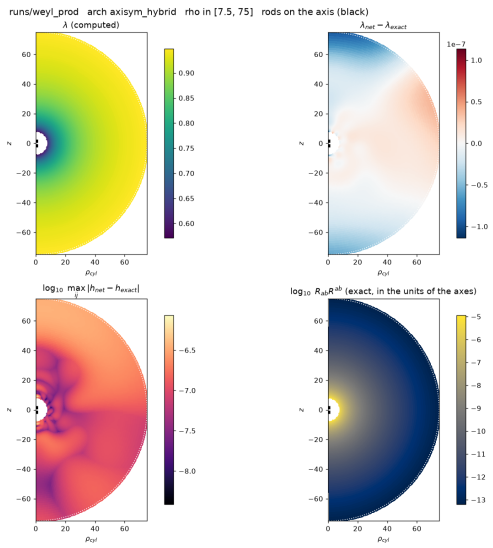


FIG. 1. Equal-mass Weyl rods on the shell $[7.5, 75]$ (meridional plane): computed λ; $\lambda_{\rm net} - \lambda_{\rm exact}$; $\log_{10} \max_{ij} |h_{\rm net} - h_{\rm exact}|$; $\log_{10} R_{ab}R^{ab}$ of the exact solution. [TODO: regenerate without the internal title]

isation, reaches a loss of 5.1×10^{-9} in 7000 iterations (17 minutes), with maximum errors 1.6×10^{-4} in h and 1.7×10^{-5} in λ, against 7.3×10^{-5} and 1.8×10^{-6} for the axisymmetric network on the same shell.

The general network is about two times less accurate than the axisymmetric one in h and ten times in λ for the same problem; it has to represent a function of three variables instead of two. The number of collocation points matters more for it: the loss is a mean over the sampled points, and with too few of them the optimiser can lower the mean while the residual grows at points it does not sample. We use 8192 points here; an earlier run of the first-order network on 2048 points showed exactly this, its loss falling while every error grew.

V. QUADRUPOLAR DATA

Having tested the solver on exact solutions, we now apply it to a problem without one: spherical data with a quadrupolar perturbation.

A. The problem

We take $R_0 = 1/\sqrt{3}$, the inner sphere $\rho_{\rm in} = 1$ with $r_{\rm in} = 1$ (so $\lambda_0 = 1/3$), $\rho_{\rm out} = 100$, and

$$\lambda|_{\rho_{\rm in}} = \tfrac{1}{3} - \tfrac{1}{12}\, P_2(\cos\theta), \tag{14}$$

a 25% perturbation of the spherical value, with no known closed-form solution. Linear theory predicts a pure $l = 2$ response decaying like ρ^{-3}, no $l = 1$ part and an $l = 0$ back-reaction of order S_2^2. Against the spherical solution the perturbation has an rms size of 2.0×10^{-2} in λ and 2.2×10^{-2} in h near the inner sphere, and 2.6×10^{-4} and 3.5×10^{-6} for $30 < \rho < 100$. We grade against the spectral reference.

B. Without and with the centre-of-mass condition

Two runs with identical settings, the formulation of Sec. II D, except for the centre-of-mass term (Appendix A), show what the condition does (Table III, Fig. 2). Without it, the error is dominated by a part that is odd under $z \to -z$ (an $l = 1$ coefficient of λ peaking at 1.4×10^{-4} near $\rho \approx 2$), which the even data forbid; every term of the loss is exactly reflection symmetric, so nothing forces this part: it is the translation mode of Sec. II E. The flux (11) confirms it: it is the same on the spheres $\rho = 1, 2, 10$ and 100 but not zero, $d_z = -1.063 \times 10^{-3}$, and the λ error near the inner sphere is $0.061\,|d_z|$. With $d = 0$ imposed on the spheres $\rho = 1, 1.5, 3, 10, 30, 100$, the error falls by a factor 3900 in λ and 2700 in h, to 9.1×10^{-9} and 1.5×10^{-7} (rms over the shell), six and five orders below the size of the perturbation. The loss does not see the difference: it is 1.6×10^{-11} without the condition and 1.3×10^{-11} with it, because the translation costs almost nothing in it. For the same reason its size without the condition is not fixed: it depends on how far the optimisation happens to drift along the mode (a run with the additional outer conditions of Sec. V C and otherwise the same settings drifted to $d_z = 7.6 \times 10^{-5}$, 14 times less). The condition removes it.

C. Outer conditions that are not needed

The run with the centre-of-mass condition above uses only the pointwise outer conditions of Sec. II D (the last row of Table IV). The same run from scratch (Adam and two SSBroyden stages) with the extra outer spheres, with the averaged outer conditions, or with both is no better (Table IV): with both, the loss is seven times higher and the error three times higher in λ and in h. The only quantity they improve is the far-field λ error, from 7×10^{-9} to 9×10^{-10}, which is already 4×10^4 times below the perturbation there. The extra conditions were compensating for the over-determined inner condition; once that is removed they only add terms that compete with the equations.

D. Off-axis data

The same datum about the axis $(1, 0, 1)/\sqrt{2}$ is a fully three-dimensional problem whose solution is the rotated axisymmetric one. We solve it with the general metric-only network in the formulation of Sec. II D, Adam and four SSBroyden stages. At width 20 (2347 parameters, 16384 points) the loss levels off at 1.2×10^{-6}, five orders

TABLE III. Error against the spectral reference (angular rms) by radial region, for the same run without and with the centre-of-mass condition (Adam 5000 steps, then 2 × 10000 SSBroyden iterations; the loss is the final training loss).

	loss	λ error				h error			
		1–3	3–10	10–30	30–100	1–3	3–10	10–30	30–100
perturbation size		2.0×10^{-2}	2.5×10^{-3}	7.6×10^{-4}	2.6×10^{-4}	2.2×10^{-2}	4.5×10^{-4}	3.1×10^{-5}	3.5×10^{-6}
without centre of mass	1.6×10^{-11}	6.5×10^{-5}	3.2×10^{-5}	4.9×10^{-6}	5.9×10^{-7}	8.2×10^{-4}	1.2×10^{-4}	1.2×10^{-5}	1.2×10^{-6}
with centre of mass	1.3×10^{-11}	1.2×10^{-8}	8.9×10^{-9}	7.5×10^{-9}	7.0×10^{-9}	3.0×10^{-7}	3.4×10^{-8}	4.9×10^{-9}	4.2×10^{-9}

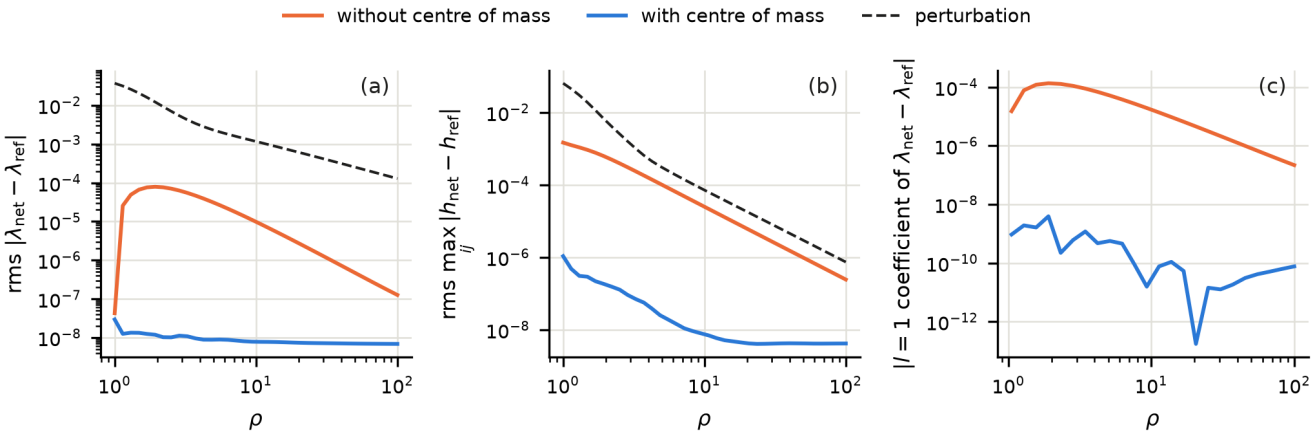


FIG. 2. Error against the spectral reference without and with the centre-of-mass condition: angular rms of $\lambda_{\rm net} - \lambda_{\rm ref}$, of $\max_{ij} |h_{\rm net} - h_{\rm ref}|$, and the $l = 1$ coefficient of $\lambda_{\rm net} - \lambda_{\rm ref}$. Dashed: the perturbation (reference minus spherical solution).

TABLE IV. The run with the centre-of-mass condition, with and without the additional outer conditions: final loss, ratio of the loss on fresh samples to the training loss, and rms error against the spectral reference over the shell and for $1 \le \rho \le 3$.

outer conditions	loss	fresh	λ	h	λ, 1–3	h, 1–3
all	9.2×10^{-11}	1.01	3.1×10^{-8}	4.2×10^{-7}	6.4×10^{-8}	8.6×10^{-7}
no extra spheres	3.7×10^{-11}	1.00	6.9×10^{-9}	1.8×10^{-7}	1.4×10^{-8}	3.7×10^{-7}
no averaged ones	1.9×10^{-11}	1.04	1.0×10^{-8}	1.9×10^{-7}	2.0×10^{-8}	3.9×10^{-7}
pointwise only	1.3×10^{-11}	0.99	9.1×10^{-9}	1.5×10^{-7}	1.3×10^{-8}	2.9×10^{-7}

above the axisymmetric one, decreasing by only 10–25% per stage; the same run with the additional outer conditions stops at the same level (1.35×10^{-6}), so it is the network, not the formulation, that limits it. At width 40 (32768 points) the loss falls to 4.3×10^{-9} within two stages and stays there (fresh/training ratio 1.00), and the error against the rotated reference is 3.4×10^{-6} in λ and 2.6×10^{-4} in h (rms over the shell), i.e. 2800 and 40 times below the perturbation (width 20: 9.1×10^{-6} and 1.2×10^{-3}). The error concentrates in h near the inner sphere (5.3×10^{-4} for $1 \le \rho \le 3$, against a perturbation of 2.3×10^{-2}) and falls to 3×10^{-8} beyond $\rho = 30$; $|d| \le 7.5 \times 10^{-7}$ on every sphere. The rotated Weyl test of Sec. IV reached 5×10^{-9} already at width 20, but in a much weaker field. Each stage takes 30 minutes at width 40 (10 at width 20).

a. The inner data in H^1. Since the error of every run concentrates at the inner sphere, a natural extension is to impose the inner data together with their derivatives along the sphere: we add to the loss $\rho\,(I - n \otimes n)\nabla$ of the λ residual and of the induced-metric residual $PhP - (r_{\rm in}/\rho)^2 P$ of Eq. (8), with the weight of the inner term, so that the inner condition is imposed in the H^1 norm of the sphere rather than in L^2. On the sphere PhP is the induced metric h_{AB} alone, so the new terms involve only the data, and h_{rr}, h_{rA} remain free. On the axisymmetric problem, with the same budget, it lowers the loss from 1.3×10^{-11} to 9.4×10^{-12} and the error from 9.1×10^{-9} to 5.9×10^{-9} in λ and from 1.5×10^{-7} to 1.2×10^{-7} in h (by a factor 1.8 in λ for $1 \le \rho \le 3$). On the off-axis problem at width 40 it slows the training down: after the same four stages the field-equation residuals are seven times larger and the error is 1.6–1.8 times larger (6.1×10^{-6} in λ, 4.0×10^{-4} in h), with the loss still decreasing (2.9×10^{-8}) where the run without the term had levelled off. Two further stages bring it to 1.9×10^{-8}, now levelling off at 4.5 times the loss of the run without the term, with the error still 1.4–1.5 times larger (4.7×10^{-6} in λ, 3.8×10^{-4} in h). In three dimensions the term therefore costs more than it gains, and we do not include it in the formulation.

VI. MULTIPOLE MOMENTS

a. Extraction from the axis. For a static axisymmetric vacuum solution the Geroch–Hansen moments [7, 8, 27] follow from the potential on the symmetry axis in canonical Weyl coordinates, $g = -e^{2U}\mathrm{d}t^2 + e^{-2U}[e^{2k}(\mathrm{d}W^2 + \mathrm{d}Z^2) + W^2\mathrm{d}\phi^2]$: with the static Ernst potential $\xi = -\tanh U$ on the axis, $\xi(0,Z) = \sum_j m_j/|Z|^{j+1}$, the moments are $M_j = m_j$ for $j \le 3$ and $M_4 = m_4 - \frac{1}{7}m_0(m_0m_2 - m_1^2)$ [28]; equivalently, from $U(0,Z) = \sum_r U^{(r)}/|Z|^{r+1}$, $M_0 = -U^{(0)}$, $M_1 = -U^{(1)}$, $M_2 = \frac{1}{3}(U^{(0)})^3 - U^{(2)}$, $M_3 = (U^{(0)})^2U^{(1)} - U^{(3)}$ [29]. Weyl coordinates need not be built, because only the axis is used: in our variables $U = \frac{1}{2}\ln\lambda$ and $W = |\partial_\phi|_h$; on the part of the axis that reaches infinity $k = 0$, so there $\mathrm{d}Z$ is the proper h-length along the axis; the inner ball cuts the axis in two, and the offset between the branches is the change of Z along a meridian semicircle, $\mathrm{d}Z = \star\mathrm{d}W$ in the meridian metric. We fit ξ on both branches and fix the origin of Z by mass centring ($M_1 = 0$); the error bar is the spread over fit windows starting at 1–1.5 ρ_{in} and fit orders 6 and 8. On exact solutions this returns: Schwarzschild in harmonic coordinates, $M_0 = R_0$ to 10^{-12} and M_l below 10^{-11} for $1 \le l \le 4$; the equal Weyl rods, $M_0 = 2$ and $M_2 = 2.500000$; the unequal rods, the closed-form moments to 10^{-5} relative and the centre of mass $z = 1.309091$; Curzon, $M_2 = -M^3/3$ and $M_4 = 19M^5/105$, although U itself has no quadrupole.

b. The Weyl networks. The trained solutions give $M_0 = 2.000001$ and $M_2 = 2.49950$ for the equal rods (exact 2 and 2.5), and $M_0 = 1.100000$, $M_2 = 0.29105$, $M_3 = -0.5016$ with the centre of mass at $z = 1.3091$ for the unequal rods (exact 1.1, 0.290909, −0.49983, 1.309091). For the rotated rods, extracted about the rotated axis, the three-dimensional network gives (in the chart of the run) $M_0 = 0.0314289$, $M_2 = 6.792\times10^{-6}$, $M_3 = -3.26\times10^{-7}$ and the centre of mass at 0.03740, against 0.0314286, 6.785×10^{-6}, -3.33×10^{-7} and 0.03740.

c. The quadrupolar solution. Table V lists the moments. The data raise the mass from the background value $R_0 = 0.57735$ to $M_0 = 0.583847$, a nonlinear back-reaction of 1.1% in the monopole, and the quadrupole moment is $M_2 = 0.1747$. The run without the centre-of-mass condition, whose pointwise error is 3900 times larger, has moments as close to the reference as the run with it (within 2×10^{-7} in M_2 and 4×10^{-7} in M_4): moments are geometric, and the translation mode moves the chart, not the geometry. The extraction shows the mode directly: the centre of mass of that run, measured from the centre of the coordinate sphere of radius ρ, is -8×10^{-7} at $\rho = 1$, -8.1×10^{-4} at $\rho = 2$, -1.63×10^{-3} at 10 and -1.79×10^{-3} at 70: it vanishes at the inner sphere, as the mode must, and tends to the translation $a = d_z/M_0 = -1.82\times10^{-3}$. With the condition it is below 6×10^{-8} at every radius.

d. Decay of the multipoles. Expanding $\mathcal{P}$ in Legendre modes $c_l(\rho)$ about the solution's own axis, the local power $\mathrm{d}\ln|c_l|/\mathrm{d}\ln\rho$ of the reference is −1.00, −3.00 and −5.0 for $l = 0, 2, 4$ (the last out to $\rho \approx 30$, where $c_4 \sim 10^{-10}$), and its odd modes are at 10^{-16}. The network with the centre-of-mass condition gives the same powers, with its odd modes below 10^{-8}. The three-dimensional network on the rotated data, expanded about the data axis, has odd modes of 1.3×10^{-6} at width 20, with its $l = 2$ tail holding out to $\rho \approx 30$ and its $l = 4$ tail only to $\rho \approx 5$ (M_4 good to 5%); at width 40 the odd modes are below 5×10^{-8}, the $l = 2$ tail holds to ρ_{out} and the $l = 4$ tail to $\rho \approx 20$, and M_0, M_2, M_4 are good to 10^{-6}, 2×10^{-5} and 2×10^{-3} relative. In the mass-centred harmonic chart $c_l\rho^{l+1}$ tends to 0.58387, 0.1747 and 0.0220, i.e. to M_0, M_2 and M_4, consistently with harmonic coordinates being asymptotically Cartesian and mass centred [30, 31]: a second, independent route to the same numbers.

TABLE V. Geroch–Hansen moments of the quadrupolar solution: the spectral reference, and the differences network − reference with the same fit windows (the reference's window spread, 1.2×10^{-4} in M_2 and 7×10^{-4} in M_4, largely cancels in the differences). For the rotated data the moments are taken about the data axis.

	M_0	M_2	M_4
reference	0.583847	0.17471	0.022
with centre of mass	-6.3×10^{-8}	$+7.3\times10^{-7}$	$-3.2\times$
without centre of mass	-2.5×10^{-8}	$+1.8\times10^{-7}$	$-4.1\times$
3-D network (width 20), rotated data	-5.9×10^{-6}	$+2.5\times10^{-5}$	$-1.2\times$
3-D network (width 40), rotated data	-5×10^{-7}	$+4.0\times10^{-6}$	$-3.9\times$

VII. CONCLUSIONS

A PINN solves the exterior boundary-value problem of the static vacuum Einstein equations with prescribed inner data and asymptotic flatness, and reproduces exact solutions with and without symmetry. Two ingredients are essential: a network whose connection is derived from its metric, and a complete gauge fixing. Harmonic coordinates and the inner data leave the asymptotic chart free to translate; the mode is invisible to every residual except at a very low level, and it is the largest error until the pseudo-Newtonian centre of mass is fixed. With that, the boundary conditions are minimal: λ and the induced metric on the inner sphere, pointwise Robin conditions on the outer one, nothing on h_{rr} or h_{rA}; conditions added beyond these only make the solution worse. We expect the same issue, and the same remedy, in any elliptic formulation of the stationary problem whose inner data do not fix the asymptotic frame. The Geroch–Hansen moments of the computed solutions can be read off the symmetry axis without constructing Weyl coordinates, and they are insensitive to that translation. The extension to the stationary equations is the natural next step.

Appendix A: Reproducing the results

[TODO: repository URL and the commit the paper's numbers come from.] The code is the `Stationary` package of that repository: `stationary/train.py` (the solver; every option below is a flag of `python -m stationary.train`), `stationary/specref.py` (the spectral reference), `stationary/gh_moments.py` (the moments). Software: Python 3.14, JAX 0.11.1, Flax 0.12.9, Optax 0.2.8, in double precision (`JAX_ENABLE_X64=1`); hardware: NVIDIA A30 (24 GB). Each result is produced by the commands below, all in the repository.

- Exact-solution checks: `tests/test_pipeline.py`, `verify_weyl.py`, `tests/test_gh_moments.py`.
- Schwarzschild tests (Table I): `Paper/commands_schwarzschild.sh`.
- Weyl tests: the configurations stored with each run (`runs/weyl_*/config.json`); the rotated test, `Paper/commands_weyl.sh`.
- Quadrupolar data: `recipe_pq16k.sh` (the problem). With and without the centre of mass (Table III, Fig. 2), `ab_none_*` in `Paper/hub_qp.sh` and `fn_nocom_*` in `Paper/hub_qp5.sh` (Adam 3 minutes, 10–11 minutes per SSBroyden stage on one A30). The outer conditions (Table IV), `ab_*` in `Paper/hub_qp.sh`, and `qp_com_*`: its Adam stage in `mendieta_submit.sh` (round P), its SSBroyden stages in `Paper/hub_qp.sh` (20 minutes each with the extra outer spheres). Off-axis data, `Paper/hub_qp3.sh` (`qp_rot3_*`, width 20, 10 minutes per stage) and `Paper/hub_qp4.sh` (`qp_rot4_n32768_*`, width 40, 30 minutes per stage).
- The spectral reference: `Paper/commands_specref.sh` (reproduces the loss 2.793×10^{-11}; 36 minutes on one A30).
- Grading and moments: `scripts/grade_rotated.py`, `scripts/fresh_sample_loss.py`, `scripts/gh_moments_runs.py`, `scripts/tail_decay.py`.

ACKNOWLEDGMENTS

We thank the Palma group [TODO: names] for the Crunch implementation of SSBroyden, and the CCAD of the Universidad Nacional de Córdoba for computing time on Mendieta and its JupyterHub. [TODO: funding]. The initial setting and the exact-solution tests were done with the help of DeepSeek; the quadrupole runs and the centre-of-mass fix were done with Claude (Anthropic).

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